

Lecture 3: Conditional Expectation & Martingales

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1 A Short Review of Probability Theory

We consider the **probability space** associated with the experiment of tossing a coin N times, $(\Omega, \mathcal{F}, P)$. Ω is called the sample space, which contains all possible outcomes of this experiment (all strings of N letters consisting of just “ H ” and “ T ”). An event is a subset of Ω . P is the probability measure, which is used to measure how likely an event (a subset of Ω) will occur. For example, consider an experiment of tossing a coin 3 times, and the probability of “ H ” is p , then

$$\begin{aligned} P(\{\text{the second toss results in a } H\}) &= P(\{HHH, HHT, THH, THT\}) = p^3 + 2p^2q + q^2p = p, \\ P(\{\text{two } H\text{'s and one } T\}) &= P(\{HHT, HTH, THH\}) = 3p^2q. \end{aligned}$$

The collection of all subsets¹ is denoted by $\mathcal{F}$. It is obviously a σ -algebra. (closed under complement and countable union)

A **random variable** is a real-valued function defined on Ω . In a N -period binomial model, the stock price S_k , $1 \leq k \leq N$, is a random variable. However, S_k is determined if we know the outcome of the first k tosses. In other words, the **information** available to us at time k can determine the stock price at time k . In general, we denote the collection of sets determined by the first k tosses by $\mathcal{F}_k$. It is easy to check that $\mathcal{F}_k$ is a σ -algebra, and $\mathcal{F}_k \subset \mathcal{F}_{k+1}$ (information is aggregating with time). The collection $\{\mathcal{F}_k\}$ is said to be a filtration on the given probability space.

Example 1. In the experiment of tossing a coin 3 times, the collection $\mathcal{F}_1$ of sets determined by the first toss consists of:

$$\{HHH, HHT, HTH, HTT\}, \{THH, THT, TTH, TTT\}, \phi, \Omega.$$

The collection $\mathcal{F}_2$ of sets determined by the first two tosses consists of:

$$\{HHH, HHT\}, \{HTH, HTT\}, \{THH, THT\}, \{TTH, TTT\}, \phi, \Omega,$$

and any sets that can be obtained by complements and unions of the above.

The collection $\mathcal{F}_3$ of sets determined by all three tosses coincides with $\mathcal{F}$.

An element of $\mathcal{F}_k$ is said to be $\mathcal{F}_k$ -measurable. Similarly, if a random variable X can be determined from the first k tosses, then we also say that X is $\mathcal{F}_k$ -measurable. In the N -period binomial model, S_k is $\mathcal{F}_k$ -measurable.

A **stochastic process** $X = \{X_k\}$ is said to be adapted to the filtration $\{\mathcal{F}_k\}$ if X_k is a $\mathcal{F}_k$ -measurable random variable for all $1 \leq k \leq N$. For example, in the N -period binomial model, the stock price $S = \{S_k\}$ is such a stochastic process.

Under probability measure P , we can compute the expectation of a given random variable. For example, under the physical measure, the stock price at time one in a binomial model is given by

$$E(S_1) = p \cdot uS_0 + q \cdot dS_0. \quad (1)$$

¹In general, we can consider a probability measure that is only defined for some subsets of Ω , say, only define it on $\{\phi, \Omega\}$. But we will use a probability measure that is defined for all subsets of Ω .

Similarly, we can computer the conditional expectation of a random variable. For example, the stock price S_{k+1} is $\mathcal{F}_{k+1}$ -measurable, we can make a predication of it based on $\mathcal{F}_k$,

$$E(S_{k+1}|\mathcal{F}_k) = p \cdot uS_k + q \cdot dS_k. \quad (2)$$

We notice the similarity of (1) and (2). In other words, conditional expectation is just an expectation based on knowledge of the future², if S_k turns out to be s , then at time t , the predication of S_{k+1} is just $(pu + qd)s$.

Conditional expectation has the linearity as expectation does. It also has some other useful properties

$$E(XY|\mathcal{F}_k) = Y \cdot E(X|\mathcal{F}_k) \quad \text{if } Y \text{ is } \mathcal{F}_k\text{-measurable,} \quad (3)$$

$$E(E(X|\mathcal{F}_m)|\mathcal{F}_n) = E(X|\mathcal{F}_n) \quad \text{if } n \leq m. \quad (4)$$

There is one kind of stochastic process that is of importance to us: martingale. A **martingale** is a adapted stochastic process $X = \{X_K\}$, such that

$$X_k = E(X_{k+1}|\mathcal{F}_k) = E(X_m|\mathcal{F}_k), \text{ for any } n < m \leq N. \quad (5)$$

Using (4) it is clearly seen that if X is a martingale, then

$$E(X_1) = E(X_2) = \dots = E(X_N). \quad (6)$$

2 Binomial Asset Pricing Model Revisited

The main results of binomial model are Δ_k replication/hedge and risk-neutral pricing. We focus on risk-neutral part here. Recall that, under a simple N -period binomial model (with common r, u, d at every step), the risk-neutral probability $\tilde{p}$ is chosen so that

$$S_k = \frac{1}{1+r}[\tilde{p}uS_k + \tilde{q}dS_k], \quad 0 \leq k \leq N-1 \quad (7)$$

then the price of a derivative with payoff V_N at time N can be valued by

$$V_k(S_k) = \frac{1}{1+r}[\tilde{p}V_{k+1}(uS_k) + \tilde{q}V_{k+1}(dS_k)], \quad 0 \leq k \leq N-1. \quad (8)$$

Now let us denote by $\tilde{E}$ the probability measure induced by $\tilde{p}$, then (7) is just

$$\tilde{E}\left(\frac{S_{k+1}}{1+r} \middle| \mathcal{F}_k\right) = S_k, \quad 0 \leq k \leq N-1 \quad (9)$$

which means that, the discounted stock price $\frac{S_k}{(1+r)^k}$, is a martingale under $\tilde{E}$. Moreover, from (8) we know that,

$$\tilde{E}\left(\frac{V_{k+1}(S_{k+1})}{1+r} \middle| \mathcal{F}_k\right) = V_k(S_k), \quad 0 \leq k \leq N-1 \quad (10)$$

which means that, the discounted derivative price $\frac{V_k(S_k)}{(1+r)^k}$, is a martingale under $\tilde{E}$.

In words, the risk-neutral measure $\tilde{E}$ is the³ measure under which the discounted stock price is a martingale. Under this measure, we have

$$E\left(\frac{V_N(S_N)}{(1+r)^N} \middle| \mathcal{F}_k\right) = \frac{V_k(S_k)}{(1+r)^k}, \quad 0 \leq k \leq N-1. \quad (11)$$

In particular, the time zero price of the derivative is

$$V_0 = E\left(\frac{V_N(S_N)}{(1+r)^N}\right). \quad (12)$$

²Expectation can also be interpreted as the conditional expectation given the trivial σ -algebra $\mathcal{F}_0 = \{\phi, \Omega\}$.

³We have a unique risk-neutral measure in binomial model. But we many have many in more complex models, e.g., a trinomial tree model.

Recall that at time N , the value of a call struck at K less the value of a put struck at K is equal to the value of a forward contract struck at K :

$$(S_N - K)^+ - (K - S_N)^- = S_N - K. \quad (13)$$

Taking conditional expectation of (13), we have

$$\tilde{E}\left(\frac{(S_N - K)^+}{(1+r)^{N-k}} \middle| \mathcal{F}_k\right) - \tilde{E}\left(\frac{(K - S_N)^-}{(1+r)^{N-k}} \middle| \mathcal{F}_k\right) = \tilde{E}\left(\frac{S_N - K}{(1+r)^{N-k}} \middle| \mathcal{F}_k\right). \quad (14)$$

If we denote by $C_k(S_k, K)$, $P_k(S_k, K)$, $F_k(S_k, K)$ the time k value of the call, the put and the forward contract, then we have the Put-Call parity at time k :

$$C_k(S_k, K) - P_k(S_k, K) = F_k(S_k, K) = S_k - \frac{K}{(1+r)^{N-k}}. \quad (15)$$

To see how to use the language of conditional expectation to show Δ_k replication/hedge, please refer to Theorem 2.4.5 in the textbook.

3 Cash Flow Evaluation

There are many securities, different from all those we have discussed before, have more than one payment before the maturity N :

- coupon-paying bonds;
- interest rate swaps involving a fix leg and a floating leg⁴;
- credit default swaps⁵;

Let us consider a N -period binomial model with $0 < d < 1 + r < u$, and with risk-neutral probability measure $\tilde{P}$. Let $C_0, C_1, \dots, C_N$ be an adapted stochastic process (C_k is $\mathcal{F}_k$ -measurable). Then the price at time k of the derivative that makes payments $C_k, \dots, C_N$ at times $k, \dots, N$, respectively, is

$$V_k = \tilde{E}\left(\sum_{i=k}^N \frac{C_i}{(1+r)^{i-k}} \middle| \mathcal{F}_k\right), \quad 0 \leq k \leq N. \quad (16)$$

To show why (16) is true, we need to construct a portfolio at time k with initial investment $X_k = V_k$ and Δ_k shares of stock, rebalance the portfolio wisely until time N , so that we can withdraw C_i at time i , $k \leq i \leq N$, then by no arbitrage V_k must be the fair price at time k . The idea is the same as before, however, since there are multiple payments, the discounted price of the derivative is no long a martingale. In fact, from (16) we know that⁶,

$$\begin{aligned} \tilde{E}(V_{k+1} | \mathcal{F}_k) &= \tilde{E}\left(\tilde{E}\left(\sum_{i=k+1}^N \frac{C_i}{(1+r)^{i-k-1}} \middle| \mathcal{F}_{k+1}\right) \middle| \mathcal{F}_k\right) = \tilde{E}\left(\sum_{i=k+1}^N \frac{C_i}{(1+r)^{i-k-1}} \middle| \mathcal{F}_k\right) \\ &= (1+r)\tilde{E}\left(\sum_{i=k+1}^N \frac{C_i}{(1+r)^{i-k}} \middle| \mathcal{F}_k\right) = (1+r)(V_k - C_k). \end{aligned} \quad (17)$$

Let us begin to construct the replicating portfolio at time k . At time k , we setup a portfolio with a total value of V_k , of which, we take a position Δ_k in stock, where we define

$$\Delta_i = \frac{V_{i+1}(\omega_1 \dots \omega_i H) - V_{i+1}(\omega_1 \dots \omega_i T)}{uS_i - dS_i}, \quad 0 \leq i \leq N. \quad (18)$$

⁴Party A offers B a fixed rate, and B offers A a floating rate.

⁵protection buyer offers the seller fixed rate coupon payments, the protection seller offers B a compensation if the target company defaults before the maturity of the contract.

⁶We used the property in (4).

At time k , we also withdraw C_k from our portfolio, and then we deposit the remaining $V_k - C_k - \Delta_k S_k$ into the money market. At time $k + 1$, if the $(k + 1)$ -th toss results in a H , then the portfolio's value becomes

$$X_{k+1}(\omega_1 \dots \omega_k H) = \Delta_k u S_k + (1 + r)(V_k - C_k - \Delta_k S_k). \quad (19)$$

Using (17) and (18), we have

$$\begin{aligned} X_{k+1}(\omega_1 \dots \omega_k H) &= \frac{V_{k+1}(\omega_1 \dots \omega_k H) - V_{k+1}(\omega_1 \dots \omega_k T)}{u - d} (u - 1 - r) + (1 + r)(V_k - C_k) \\ &= \tilde{q} V_{k+1}(\omega_1 \dots \omega_k H) - \tilde{q} V_{k+1}(\omega_1 \dots \omega_k T) + \tilde{p} V_{k+1}(\omega_1 \dots \omega_k H) + \tilde{q} V_{k+1}(\omega_1 \dots \omega_k T) \\ &= V_{k+1}(\omega_1 \dots \omega_k H). \end{aligned} \quad (20)$$

Similarly, if the $(k + 1)$ -th toss results in a T , we also have $X_{k+1} = V_{k+1}$. Using mathematical (forward) induction we can show that, when we withdraw C_i at time i , and hold Δ_i shares of stock, we always have

$$X_{i+1} = V_{i+1}, \quad k \leq i \leq N - 1. \quad (21)$$

This means that, starting at time k , the portfolio constructed can deliver us payment C_i at time i for $k \leq i \leq N - 1$; and at time N , the portfolio's value is

$$X_N = V_N = \tilde{E} \left(\sum_{i=N}^N \frac{C_i}{(1 + r)^{i-N}} \middle| \mathcal{F}_N \right) = C_N. \quad (22)$$

So this portfolio can fully replicate the security. By no arbitrage, (16) is the fair price.